

Alternative **Positive** for our Project

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Exercises

PList alternative to Positive

Define a data-type **PList** such that:

- it represents all the positive natural numbers
- the encoding is free of useless leading zeroes
- the intuitive general structure of the numbers is:

```
d_n:d_{n-1}:...:d_{2}:d_{1}:[]  
-- is the binary number:  
d_n d_{n-1} ... d_{2} d_{1} 1  
-- where:  
-- - [] is the MSB 1  
-- - d_i is the i-th digit such that:  
--   -- d_i == False stands for "zero"  
--   -- d_i == True  stands for "one"
```

- Complete **Numbers.Positive.Exercises.PList**
- **Numbers.Positive.Exercises.PListSolutions.hs** is restricted

PEither alternative to Positive

Define a data-type **PEither** such that:

- it represents all the positive natural numbers
- the encoding is free of useless leading zeroes
- the intuitive general structure of the numbers is:

```
-- PE abbreviates PEither, L abbreviates Left, R abbreviates Right
PE R(d_n (PE R(d_{n-1} ... (PE R(d_{2} (PE R(d_{1} (PE L ())))))))
-- or
(PE.R.d_n.PE.R.d_{n-1}. ... .PE.R.d_{2}.PE.R.d_{1}.PE.L) ()
-- is the binary number:
d_n d_{n-1} ... d_{2} d_{1} ()
-- where:
-- - () is the MSB 1
-- - d_i is the i-th digit such that:
--   -- d_i == R stands for "zero"
--   -- d_i == L stands for "one"
```

- Complete **Numbers.Positive.Exercises.PEither**
- **Numbers.Positive.Exercises.PEitherSolutions.hs** is restricted